

Review Session Notes

Tuesday, March 4, 2025

12:45 PM

4.8

$$\text{Given} - C_i(x, t) = C_i^0 + (C_i^* - C_i^0) \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$$

Derivation:

$$C_i(x, t) - C_i^0 = (C_i^* - C_i^0) \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$$

$$\frac{C_i(x, t) - C_i^0}{C_i^* - C_i^0} = \operatorname{erfc} \left[\frac{x}{2\sqrt{Dt}} \right]$$

Input Parameters:

$$\frac{0.6 - 0.3}{1 - 0.3} = \operatorname{erfc} \left[\frac{0.0003 \text{ m}}{2\sqrt{1.1 \times 10^{-9} \text{ m}^2/\text{s}} t} \right] \quad w$$

$$0.4286 = \operatorname{erfc}[w]$$

$$1 - 0.4286 = \operatorname{erf}[w]$$

↳ normal error function

$$0.5714 = \operatorname{erf}[w]$$

$$w = 0.56$$

$$0.56 = \frac{0.0003 \text{ m}}{\sqrt{1.1 \times 10^{-9} \text{ m}^2/\text{s}} t}$$

$$\sqrt{1.1 \times 10^{-9} \text{ m}^2/\text{s}} t = 0.000268 \text{ m}$$

$$1.1 \times 10^{-9} \text{ m}^2/\text{s} t = 7.175 \times 10^{-8} \text{ m}^2$$

$$t = 65 \text{ seconds}$$

4.5

$$\text{Given} - C_{H_2}(r=a, t=t_{1/2}) = C_{H_2}^* \left[1 - 2e^{-D t_{1/2} (r/a)^2} \right]$$

↳ also calculate $\sqrt{1.1 \times 10^{-9}}$ 11

Given - $C_{H_2}(r=0, t=t_{1/2}) = C_{H_2}^{\infty} [1 - 2e^{-D t_{1/2} (\pi/R)^2}]$
 $\hookrightarrow$ also equals $\frac{1}{2} C_{H_2}^{\infty}$ from problem

Derivation:

$$\frac{1}{2} C_{H_2}^{\infty} = C_{H_2}^{\infty} [1 - 2e^{-D t_{1/2} (\pi/R)^2}]$$

$$\frac{1}{2} = 1 - 2e^{-D t_{1/2} (\pi/R)^2}$$

$$\frac{1}{2} = 2e^{-D t_{1/2} (\pi/R)^2}$$

$$\frac{1}{4} = e^{-D t_{1/2} (\pi/R)^2}$$

$$\ln(\frac{1}{4}) = -D t_{1/2} (\pi/R)^2$$

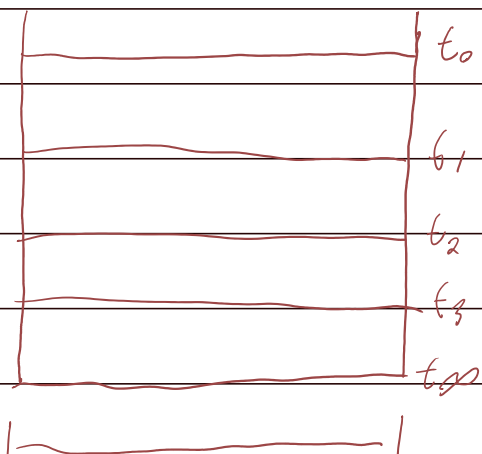
$$t_{1/2} = \frac{\ln(\frac{1}{4})}{-D (\pi/R)^2} \rightarrow \text{It says determine } t_{1/2} \text{ but in this case it just means derive this formula}$$

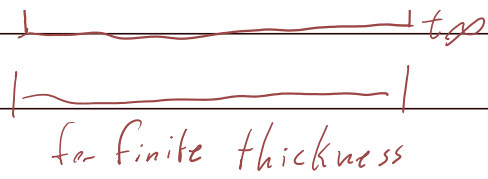
q. 10 - "Really, really important if you catch my drift"

a) Diffusion of H_2 in Solid Steel is the RDS
 - why?

• "Because its slower"

b) If RDS is H_2 desorption, there will be decreasing concentrations w/ no concentration gradients

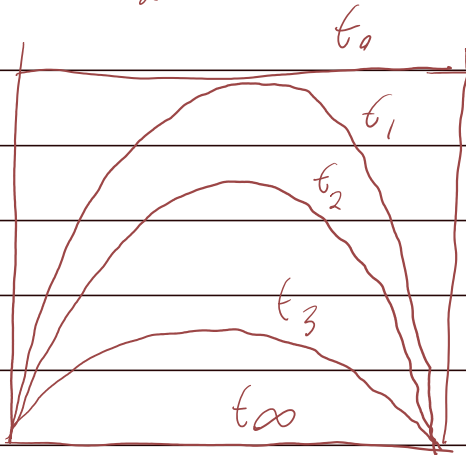




 t_{∞}

 for finite thickness

(c) If H_2 diffusion in steel is RDS



Tips/ This will be on the exam

1. True or False: If this is the generalized equation for transport:

$$J_i = \sum_K M_{ik} F_K$$

M_{ik} is the coupling coefficient between force and flux

Literally have no memory of covering this in class, but I found this in the textbooks:

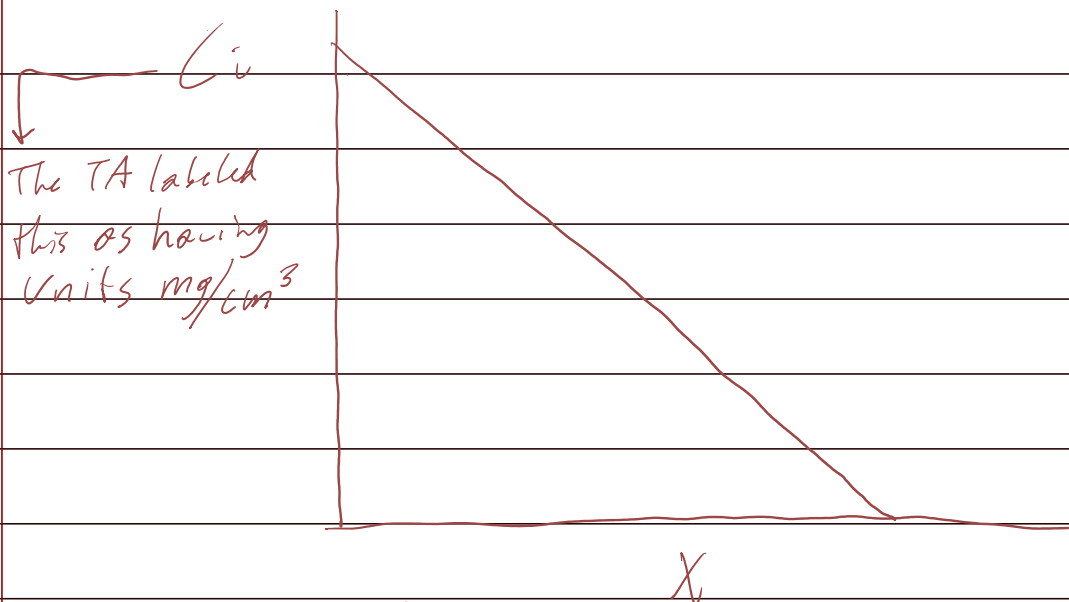
4.2 FLUXES AND FORCES

While there are many possible types of flux, a single common principle underpins all transport phenomena. This is the idea that a driving force must be present to cause the transport process to occur. If there is no driving force, there is no reason to move! This applies equally well to the transport of matter as it does to the transport of heat or charge. The governing equation for transport can be generalized (in one dimension) as

$$J_i = \sum_k M_{ik} F_k \quad (4.7)$$

where J_i represents a flux of species i , F_k represent the k different forces acting on i , and the M_{ik} 's are the coupling coefficients between force and flux. The coupling coefficients quantify the relative ability of a species to respond to a given force with movement as well as the effective strength of the driving force itself. The coupling coefficients are therefore a property both of the species that is moving and the material through which it is moving. This general equation is valid for any type of transport (charge, heat, mass, etc.) and can be used to capture the coupled effect of multiple driving forces acting simultaneously.

2. If diffusivity does not vary with the concentration and steady-state diffusion occurs, what does the concentration profile look like?



From Fick's First Law

3. Important to learn BC and IC for solid state diffusion and semi-infinite medium scenarios.